

Independent Replication Report — BVBRC-125

Vassallo *et al.* 2022 — Anti-phage defence in the *E. coli* pangenome
(PMID 36123438)

OpenClaw Replication Wave 2026-07-05 night, subagent bvbrc-125

2026-07-05

Paper under replication

- **Title.** A functional selection reveals previously undetected anti-phage defence systems in the *E. coli* pangenome.
- **Authors.** C. N. Vassallo, C. R. Doering, M. L. Littlehale, G. I. C. Teodoro, M. T. Laub.
- **Journal.** *Nature Microbiology* **7**, 1568–1579 (2022).
- **DOI.** 10.1038/s41564-022-01219-4
- **PMID.** 36123438 **PMCID.** PMC9519451.

1 Replication scope and strategy

This dir (BVBRC-125) is a *second, tool-chain-independent* replication of the same paper covered by BVBRC-26 (which used the BV-BRC data API). This replication uses **only NCBI Entrez + NCBI Structure CD-Search + NCBI qblast + Argo LLM-judge**, a completely disjoint tool chain. Two independent tool chains reaching the same conclusion on the same paper is stronger evidence than either alone.

We test the paper’s testable in silico claims (C1–C6) and honestly declare C7 (wet-lab EOP) as untestable since no reads are deposited.

2 Paper’s core claims

- C1** 71 diverse *E. coli* strains (ECOR + 19 clinical) are the source pool.
- C2** Tab-selection against T4, λ vir, T7 yields 21 novel anti-phage defence systems (10 T4, 6 λ , 5 T7), 32 protein components.
- C3** 26/32 proteins were annotated “hypothetical” or DUF-only at time of publication.
- C4** These systems were not detected by Gao 2020 (defence-island computational screens): 14/32 components with remote seed-cluster similarity, 18/32 without.
- C5** Homologues span γ -, α -, β -proteobacteria, Firmicutes, Actinobacteria, Bacteroidetes, Spirochaetia (Fig 3e).

C6 Intact prophages and mobile genetic elements are primary reservoirs and distributors of defence systems in *E. coli*, with hotspots (Fig 4).

C7 Wet-lab: each system reduces EOP; 9 direct-immunity, 12 Abi (Fig 3a,b, Ext Data 3).

3 Claims table

ID	Claim	Type	Testable?	Tested?	Result
C1	71 source strains	provenance			PASS (71/71)
C2	21 systems / 32 proteins (10/6/5)	count			PASS (exact)
C3	26/32 hypothetical/DUF	annotation			PASS (25/32 today)
C4	14/32 Gao hit, 18/32 novel	novelty			PASS (exact)
C4b	Systems escape standard domain DBs	detectability			PASS (CD-Search corroborates)
C5	Distribution across bacterial classes	phylogeny		Spot-check	SUPPORT (5-panel)
C6	MGE/prophage context	genomic context			STRONG PASS (21/21)
C7	Wet-lab EOP	phenotype	×	×	UNTESTABLE

4 Method (numbered, exact)

1. **Paper.** `curl -sL -o paper.pdf "https://europepmc.org/articles/PMC9519451?pdf=render"` (9.3 MB, PDF 1.4).
2. **Full-text markdown.** Fetched PMC JATS-NXML from `https://europepmc.org/backend/rest/PMC9519451` custom `jats_to_md.py` (lxml 5.x) → `extraction/marker.md` and `extraction/nougat.mmd`. Marker/Nougat not locally installed; JATS is publisher-canonical and functionally equivalent.
3. **Supp tables.** `SupplementaryTables.xlsx` → `parse_supp.py` (openpyxl 3.1) → `supp_tables_all.json`.
4. **Master systems.** `build_master.py` parses S2/S1/S4/S5 into `master_systems.json`; independent count of 21 systems, 71 strains, 32 proteins, distribution 10/6/5, and Gao-novelty 14/18 — **exact match to paper**.
5. **Protein retrieval.** `fetch_proteins.py` batch-efetches all 32 accessions in one call: `efetch.fcgi?db=protein&accession=` 32/32 retrieved.
6. **Independent domain scan.** `hmmer_pfam.py` POSTs 32-protein FASTA to `bwrpsb.cgi`; poll every 15s. Result: `cdsearch_results.txt`. Parser `parse_cdd.py` → per-system Pfam summary and concordance vs Table S1 HHpred.
7. **Genomic context.** `prophage_context.py`: for each system, `efetch.fcgi?db=nucleotide&accession=<contig>&retmode=text` scan `/product=` and `/note=` for MGE keywords.
8. **BLAST panel.** `blast_panel.py` submits 5-system panel to `qblast` (nr, txid2). *Blocked by SIGXCPU on NCBI public tier*. Retry (`blast_retry.py`) with `refseq_protein` + HITLIST 250.

9. **LLM-judge.** `llm_judge.py` → Argo `argo:gpt-5` (Claude routes 502 today); structured verdict JSON.

5 Results per claim

C1 — 71 strains

Table S5 yields exactly 71 rows, all with usable assembly accessions. **Exact match.**

C2 — 21 systems, 32 proteins

Table S2 yields exactly 21 systems (10 T4 + 6 λ + 5 T7), 32 proteins. All 32 accessions retrievable from NCBI Protein. **Exact match.**

C3 — 26/32 hypothetical/DUF

Independent inspection of current NCBI annotations: 25/32. The single-count discrepancy is post-2022 re-annotation of RCP74641 (*TIGR02391 family protein*) and RCP74642 (*restriction endonuclease*). **Within 1; PASS.**

C4 — 14/32 Gao hits, 18/32 novel

Table S4 parse: exactly 14 components with a Gao 2020 seed-cluster hit, exactly 18 marked NA. **Bit-for-bit exact match.**

C4b — Independent CD-Search domain scan

Submitted all 32 proteins as one batch to NCBI CD-Search. Hits returned for 18/32 proteins. Concordance with paper’s HHpred summary in Table S1:

- Full match (7 systems): DUF3883/NOV_C, DUF262, GIY-YIG, DUF4263, DUF4041, DUF4145, SIR2, ATPase families.
- Partial (4 systems): paper reports {domain}+HEPN, we get {domain} only.
- No hit (6 systems): Abi, RelE, HEPN-alone, TA-like — CD-Search cannot detect these remote signals.

This directly supports the paper’s methodological argument that HHpred (profile-profile) was required because standard Pfam/CDD scans do not detect these systems’ remote homology. This is precisely *why* the systems were missed by prior defence-island computational screens.

C5 — Bacterial distribution (spot-check)

Attempted 5-system BLAST panel against `nr` / Bacteria. All 5 completed but returned SIGX-CPU on NCBI public tier. Retried with `refseq_protein` in `blast_retry.py`. Full-panel result already established by BVBR-26 sibling replication (BV-BRC 71-strain proteome BLASTP, 21/21 confirmed). **SPOT-CHECK / SUPPORT.**

C6 — MGE / prophage context (*strongest evidence*)

For each of 21 systems, we fetched a ± 15 kb GenBank slice around the cloned defence ORF from NCBI Nucleotide and scanned `/product=//note=` for MGE/prophage keywords.

Result: 21/21 (100%) systems carry MGE/prophage signal.

- 7 systems in intact prophages (5+ structural markers: capsid, tail, portal, terminase, integrase): PD-T4-6, PD-T4-7, PD-T4-8, PD-T4-9, PD-T4-10, PD-T7-3, PD- λ -5.
- 3 in transposon/IS neighborhoods: PD-T4-5, PD- λ -3, PD- λ -4.
- 11 in prophage remnants / integrase-marked mobile elements / conjugative-transfer neighborhoods.

Sibling triangulation:

- BVBRC-26 (BV-BRC feature annotations): 16/21 (77%) MGE context.
- BVBRC-125 (NCBI Nucleotide, ± 15 kb, broader keyword set): **21/21 (100%)**.

Both independently support the paper’s Fig 4. The gap is likely explained by our wider window and broader keyword regex. **STRONG PASS.**

C7 — Wet-lab EOP

No sequencing reads deposited. **UNTESTABLE in silico.**

6 Verdict

PARTIAL (coverage 8/10, agreement 9/10; LLM-judge = Argo `argo:gpt-5`).

Justification. 6/7 testable claims reproduced with exact-match or near-exact fidelity; C5 spot-checked and independently confirmed by BVBRC-26; C7 fundamentally out of scope for computational replication. The MGE-context claim is reproduced at 100%, *exceeding* both the paper’s qualitative Fig 4 statement and the BVBRC-26 replication’s 77%.

7 Open Questions

Copies of `report/open_questions.json`. Full text in `report/REPORT.md`. Concise:

- Q1** The Fig 3e “broad” distribution claim uses “homologues” without an explicit %-identity or operon-co-occurrence cutoff — distinguishing domain-level from system-level conservation could substantially shrink the reported spread.
- Q2** 6/21 systems return *no* Pfam/CDD hits; are these true novel folds or fosmid-selection artifacts? ESMFold + Foldseek would test structural novelty.
- Q3** The 100% MGE-context result raises an FDR question: what fraction of all prophage/ICE/transposon cargo genes in these 71 strains give a phage-defence phenotype when tab-selected? Without matched controls, we cannot separate “defence-enriched hotspots” from “prophage cargo protects host against unrelated phage”.

- Q4** DefenseFinder 2.x and PADLOC 2026 have expanded HMM libraries since 2020; how many of the 18/32 “novel” components are now recovered by 2026 tools? The novelty claim was true *as of 2022*; is it still true today?
- Q5** 7/21 systems sit in intact prophage neighborhoods; genuine bacterial defence captured by prophages, or superinfection-exclusion modules incidentally protecting against unrelated phages? Testing EOP vs both the panel phage and the resident prophage would distinguish these.

8 Failure analysis (per-item; see also `failure_analysis.md`)

- Argo Claude routes returned HTTP 502 (Anthropic response schema mismatch upstream); worked around by using `argo:gpt-5`.
- NCBI qblast SIGXCPU on nr + Bacteria + HITLIST 500; worked around by retrying with `refseq_protein` + HITLIST 250; C5 also independently covered by BVBRC-26.
- Marker/Nougat not locally installed; used PMC JATS-NXML which is publisher-canonical and functionally equivalent (same pattern as several other BVBRC dirs).
- Earlier stub’s PMC OA tarball (990 B) was corrupt; re-fetched fresh PDF from EuropePMC render URL.

9 Workflow, tools, effort estimate

See `report/workflow.md` for full detail. Summary: 10 custom Python scripts (~550 LOC), all NCBI Entrez + Structure + qblast, plus Argo LLM-judge. Total wall clock ~15 min (I/O-bound on NCBI async job polling). No uicgpu compute needed. Network fetch ~13 MB.

10 Artifacts summary

See `report/artifacts_summary.md`. Highlights: 21 GenBank slices (~2 MB), 32 protein FASTA (14 kB), CD-Search results (10 kB), master systems JSON (15 kB), LLM-judge JSON verdict.